PG No. 25 C_{6v} $6mm$ [hexagonal]

* Wyckoff site: `1a`, site symmetry: `6mm`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `6mm`

No.	vector	center	mapping
1	[0, 0, Z]	[0, 0, z]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `2mm`

No.	vector	center	mapping
1	[X, $2X$, 0]	[0, 0, z]	`[1,-4,8,-11]`
2	[$-2X$, $-X$, 0]	[0, 0, z]	`[2,-5,7,-10]`
3	[X, $-X$, 0]	[0, 0, z]	`[3,-6,9,-12]`

Table 3: Wyckoff bond: `3c@1a`, bond symmetry: `2mm`

No.	vector	center	mapping
1	[X, 0, 0]	[0, 0, z]	`[1,-4,-8,11]`
2	[0, X, 0]	[0, 0, z]	`[2,-5,-7,10]`
3	[$-X$, $-X$, 0]	[0, 0, z]	`[3,-6,-9,12]`

Table 4: Wyckoff bond: `6d@1a`, bond symmetry: `2..`

No.	vector	center	mapping
1	[X, Y, 0]	[0, 0, z]	`[1,-4]`
2	[$-Y$, $X-Y$, 0]	[0, 0, z]	`[2,-5]`
3	[$-X+Y$, $-X$, 0]	[0, 0, z]	`[3,-6]`
4	[$-Y$, $-X$, 0]	[0, 0, z]	`[7,-10]`
5	[$-X+Y$, Y, 0]	[0, 0, z]	`[8,-11]`
6	[X, $X-Y$, 0]	[0, 0, z]	`[9,-12]`

Table 5: Wyckoff bond: `6e@1a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	[X, $2X$, Z]	[0, 0, z]	`[1,8]`
2	[$-2X$, $-X$, Z]	[0, 0, z]	`[2,7]`
3	[X, $-X$, Z]	[0, 0, z]	`[3,9]`
4	[$-X$, $-2X$, Z]	[0, 0, z]	`[4,11]`
5	[$2X$, X, Z]	[0, 0, z]	`[5,10]`
6	[$-X$, X, Z]	[0, 0, z]	`[6,12]`

Table 6: Wyckoff bond: `6f@1a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, z]$	`[1,11]`
2	$[0, X, Z]$	$[0, 0, z]$	`[2,10]`
3	$[-X, -X, Z]$	$[0, 0, z]$	`[3,12]`
4	$[-X, 0, Z]$	$[0, 0, z]$	`[4,8]`
5	$[0, -X, Z]$	$[0, 0, z]$	`[5,7]`
6	$[X, X, Z]$	$[0, 0, z]$	`[6,9]`

Table 7: Wyckoff bond: `12g@1a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, z]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, 0, z]$	`[5]`
6	$[X-Y, X, Z]$	$[0, 0, z]$	`[6]`
7	$[-Y, -X, Z]$	$[0, 0, z]$	`[7]`
8	$[-X+Y, Y, Z]$	$[0, 0, z]$	`[8]`
9	$[X, X-Y, Z]$	$[0, 0, z]$	`[9]`
10	$[Y, X, Z]$	$[0, 0, z]$	`[10]`
11	$[X-Y, -Y, Z]$	$[0, 0, z]$	`[11]`
12	$[-X, -X+Y, Z]$	$[0, 0, z]$	`[12]`

* Wyckoff site: `6b`, site symmetry: `..m`

Table 8: Wyckoff bond: `6a@6b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, z]$	`[1,11]`
2	$[0, X, Z]$	$[0, x, z]$	`[2,10]`
3	$[-X, -X, Z]$	$[-x, -x, z]$	`[3,12]`
4	$[-X, 0, Z]$	$[-x, 0, z]$	`[4,8]`
5	$[0, -X, Z]$	$[0, -x, z]$	`[5,7]`
6	$[X, X, Z]$	$[x, x, z]$	`[6,9]`

Table 9: Wyckoff bond: `6b@6b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[x, 0, z]$	`[1,-11]`
2	$[-2X, -X, 0]$	$[0, x, z]$	`[2,-10]`
3	$[X, -X, 0]$	$[-x, -x, z]$	`[3,-12]`
4	$[-X, -2X, 0]$	$[-x, 0, z]$	`[4,-8]`
5	$[2X, X, 0]$	$[0, -x, z]$	`[5,-7]`
6	$[-X, X, 0]$	$[x, x, z]$	`[6,-9]`

Table 10: Wyckoff bond: `12c@6b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, x, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, 0, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, -x, z]$	`[5]`
6	$[X-Y, X, Z]$	$[x, x, z]$	`[6]`
7	$[-Y, -X, Z]$	$[0, -x, z]$	`[7]`
8	$[-X+Y, Y, Z]$	$[-x, 0, z]$	`[8]`
9	$[X, X-Y, Z]$	$[x, x, z]$	`[9]`
10	$[Y, X, Z]$	$[0, x, z]$	`[10]`
11	$[X-Y, -Y, Z]$	$[x, 0, z]$	`[11]`
12	$[-X, -X+Y, Z]$	$[-x, -x, z]$	`[12]`

* Wyckoff site: `6c`, site symmetry: `.m.`

Table 11: Wyckoff bond: `6a@6c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, -x, z]$	`[1,7]`
2	$[X, 2X, Z]$	$[x, 2x, z]$	`[2,9]`
3	$[-2X, -X, Z]$	$[-2x, -x, z]$	`[3,8]`
4	$[-X, X, Z]$	$[-x, x, z]$	`[4,10]`
5	$[-X, -2X, Z]$	$[-x, -2x, z]$	`[5,12]`
6	$[2X, X, Z]$	$[2x, x, z]$	`[6,11]`

Table 12: Wyckoff bond: `6b@6c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, -x, z]$	`[1,-7]`
2	$[-X, 0, 0]$	$[x, 2x, z]$	`[2,-9]`
3	$[0, -X, 0]$	$[-2x, -x, z]$	`[3,-8]`
4	$[-X, -X, 0]$	$[-x, x, z]$	`[4,-10]`
5	$[X, 0, 0]$	$[-x, -2x, z]$	`[5,-12]`
6	$[0, X, 0]$	$[2x, x, z]$	`[6,-11]`

Table 13: Wyckoff bond: `12c@6c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, -x, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[x, 2x, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-2x, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, x, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[-x, -2x, z]$	`[5]`
6	$[X-Y, X, Z]$	$[2x, x, z]$	`[6]`

continued ...

Table 13

No.	vector	center	mapping
7	$[-Y,\ -X,\ Z]$	$[x,\ -x,\ z]$	[7]
8	$[-X+Y,\ Y,\ Z]$	$[-2x,\ -x,\ z]$	[8]
9	$[X,\ X-Y,\ Z]$	$[x,\ 2x,\ z]$	[9]
10	$[Y,\ X,\ Z]$	$[-x,\ x,\ z]$	[10]
11	$[X-Y,\ -Y,\ Z]$	$[2x,\ x,\ z]$	[11]
12	$[-X,\ -X+Y,\ Z]$	$[-x,\ -2x,\ z]$	[12]

* Wyckoff site: 12d, site symmetry: 1

Table 14: Wyckoff bond: 12a@12d, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$[-y,\ x-y,\ z]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$[-x+y,\ -x,\ z]$	[3]
4	$[-X,\ -Y,\ Z]$	$[-x,\ -y,\ z]$	[4]
5	$[Y,\ -X+Y,\ Z]$	$[y,\ -x+y,\ z]$	[5]
6	$[X-Y,\ X,\ Z]$	$[x-y,\ x,\ z]$	[6]
7	$[-Y,\ -X,\ Z]$	$[-y,\ -x,\ z]$	[7]
8	$[-X+Y,\ Y,\ Z]$	$[-x+y,\ y,\ z]$	[8]
9	$[X,\ X-Y,\ Z]$	$[x,\ x-y,\ z]$	[9]
10	$[Y,\ X,\ Z]$	$[y,\ x,\ z]$	[10]
11	$[X-Y,\ -Y,\ Z]$	$[x-y,\ -y,\ z]$	[11]
12	$[-X,\ -X+Y,\ Z]$	$[-x,\ -x+y,\ z]$	[12]